Understanding Agentic Specification and Generated Code

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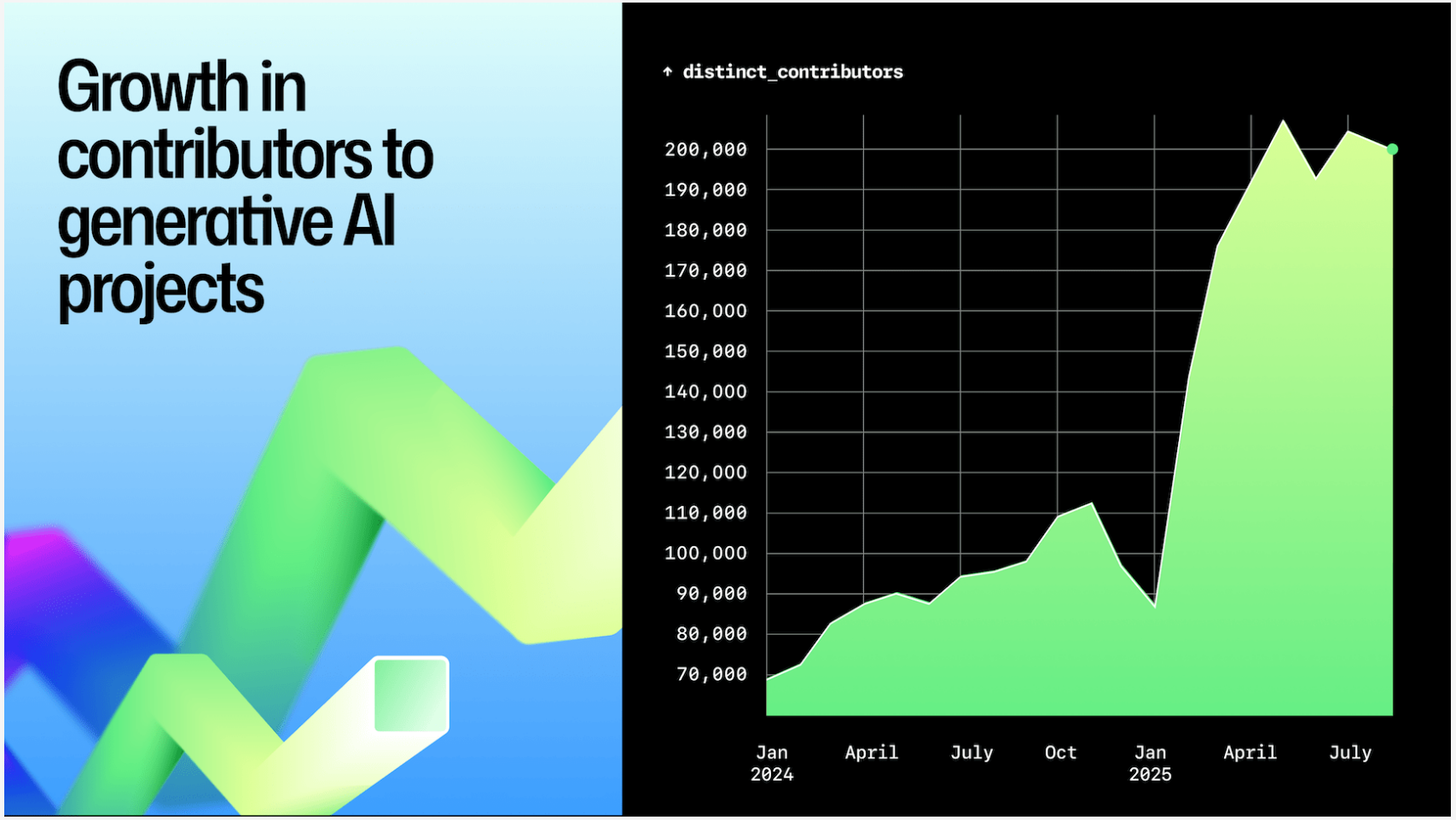
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Outline

1. **Introduction**
   - AI Agents in Software Engineering
   - Agentic context files
   - Token usage and AI slops
2. **Problem Statement**: Agentic specification for generated code
3. **Methodology**
   - Data collection process
   - Specification generation process
   - Library reconstruction process
   - Human vs. AI comparison
4. **Initial Results**
5. **Future Works**

Introduction – AI Agents in Software Engineering



Source: GitHub Octoverse 2025

Introduction – Agentic Context Files

What is a CLAUDE.md file?

Here's an example CLAUDE.md that you might have in your repository:

```
# Project Context

When working with this codebase, prioritize readability over cleve

## About This Project

FastAPI REST API for user authentication and profiles. Uses SQLAlc

## Key Directories

- `app/models/` - database models
- `app/api/` - route handlers
- `app/core/` - configuration and utilities

## Standards

- Type hints required on all functions
- pytest for testing (fixtures in `tests/conftest.py`)
- PEP 8 with 100 character lines

## Common Commands
```bash
uvicorn app.main:app --reload  # dev server
pytest tests/ -v               # run tests
```
```

AGENTS.md

A simple, open format for guiding coding agents, used by over 20k open-source projects.

```
# AGENTS.md

## Setup commands
- Install deps: `pnpm install`
- Start dev server: `pnpm dev`
- Run tests: `pnpm test`

## Code style
- TypeScript strict mode
- Single quotes, no semicolons
- Use functional patterns where possible
```

Introduction – Token Usage and AI Slops

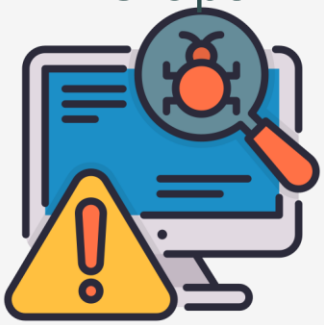
AI Token*



Token types

- Input Token
- Output Token
- Context Token
- Reasoning Token

AI Slops**

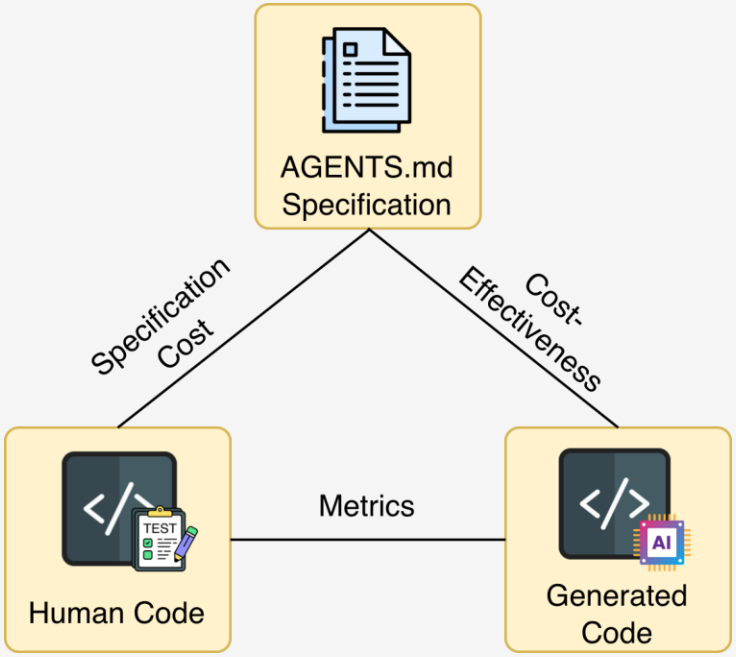


- Redundant Code
- Dead Code
- Excessive Comments
- Hallucinations

*Salim et al., *Tokenomics: Quantifying Where Tokens Are Used in Agentic Software Engineering*, 2026

**Shaib et al., *Measuring AI "Slop" in Text*, 2026

Problem Statement – Key Components

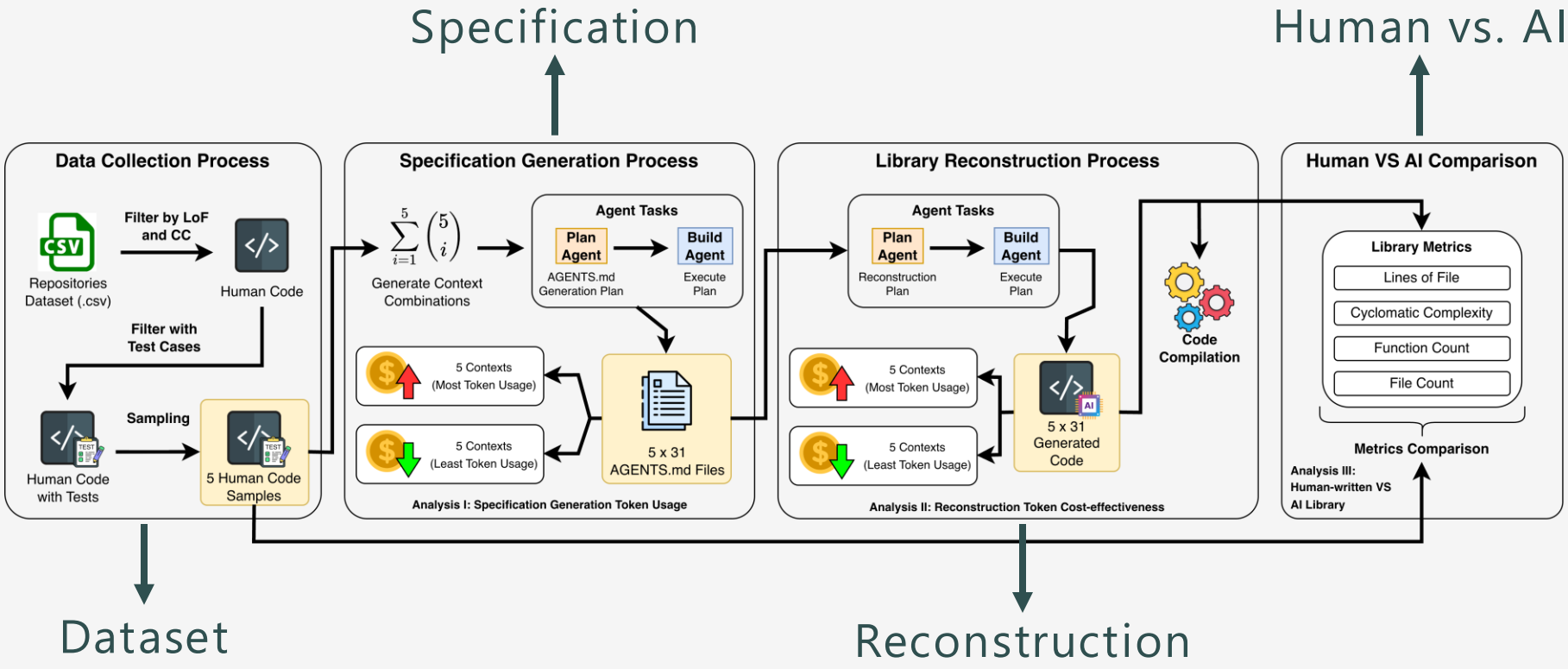


- Human Code to Agentic Specification
- Agentic Specification to AI-Generated Code
- AI-Generated Code to Human Code

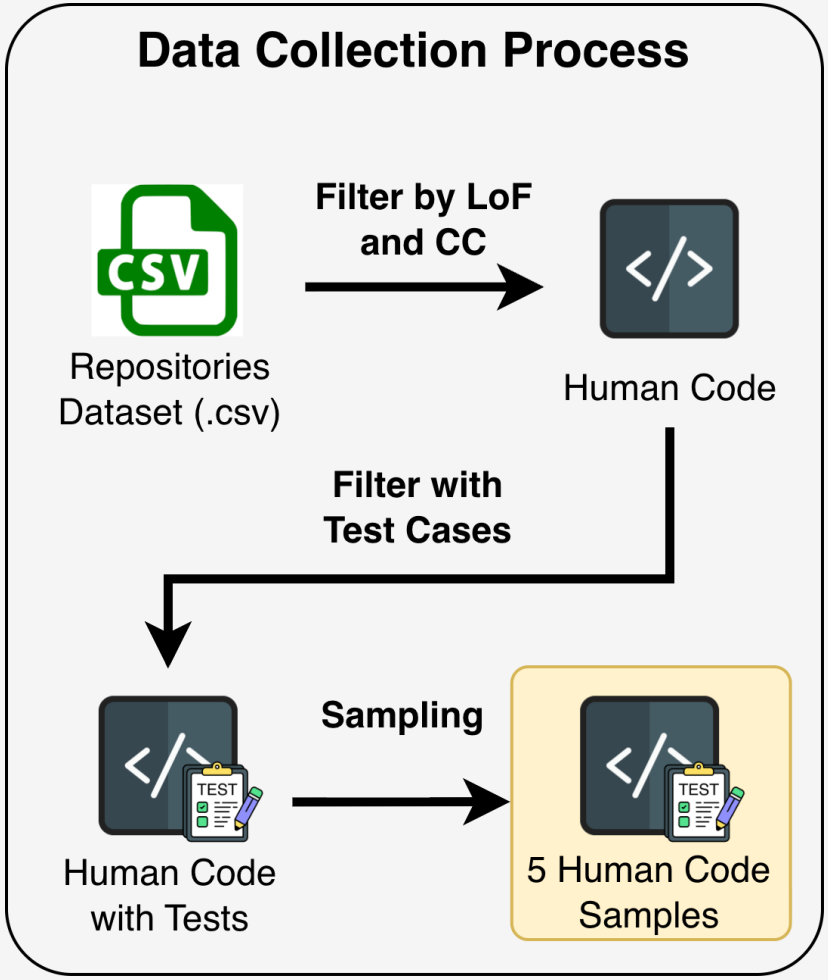
Problem Statement – Research Questions (RQ)

- RQ1: Which types of context result in the **highest** and **lowest** token usage for creating agentic specification files?
- RQ2: Which contexts **minimize** or **maximize** token usage while still generate code that can be compiled?
- RQ3: How do different specification contexts affect the **structural characteristic** of AI-generated code to the original human-written code?

Methodology – Full Pipeline



Methodology – Data Collection Process



Human Code:

PyPI Trivial Libraries*

Condition:

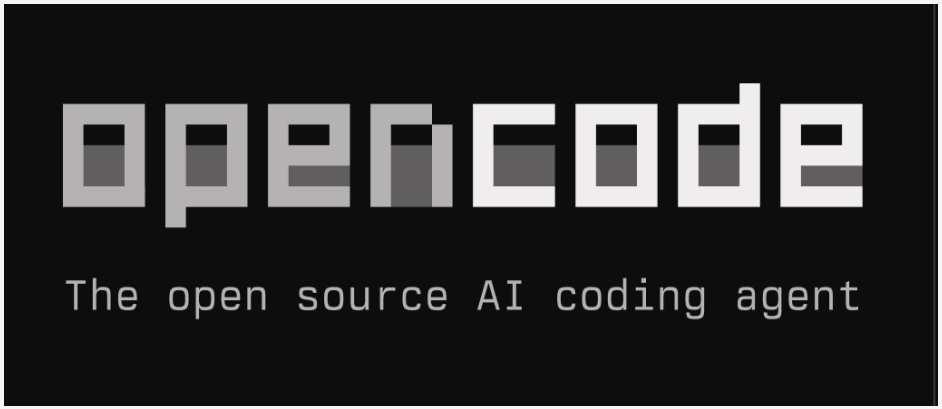
$$\mathbf{LoF} \leq \mathbf{35} \cup \mathbf{CC} \leq \mathbf{10}$$

LoF: Lines of Files

CC: Cyclomatic Complexity

*Abdalkareem et al., *On the impact of using trivial packages*, 2020

Methodology – AI Agents and LLM



OpenCode

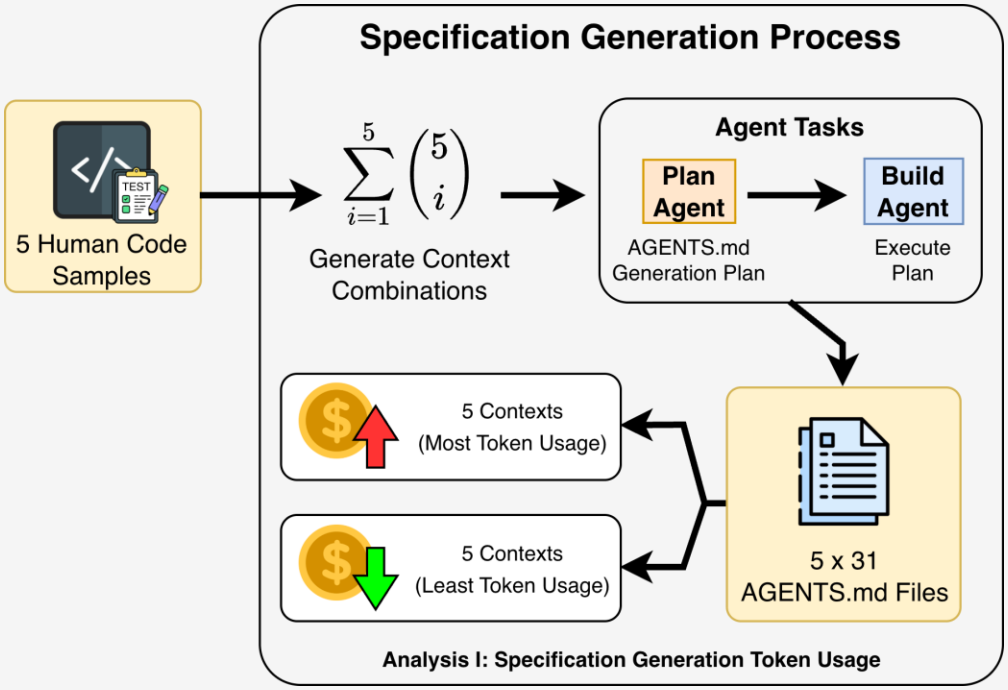


Gemma4 (Local Model)



GPT-5.4 (OpenAI API)

Methodology – Specification Generation Process

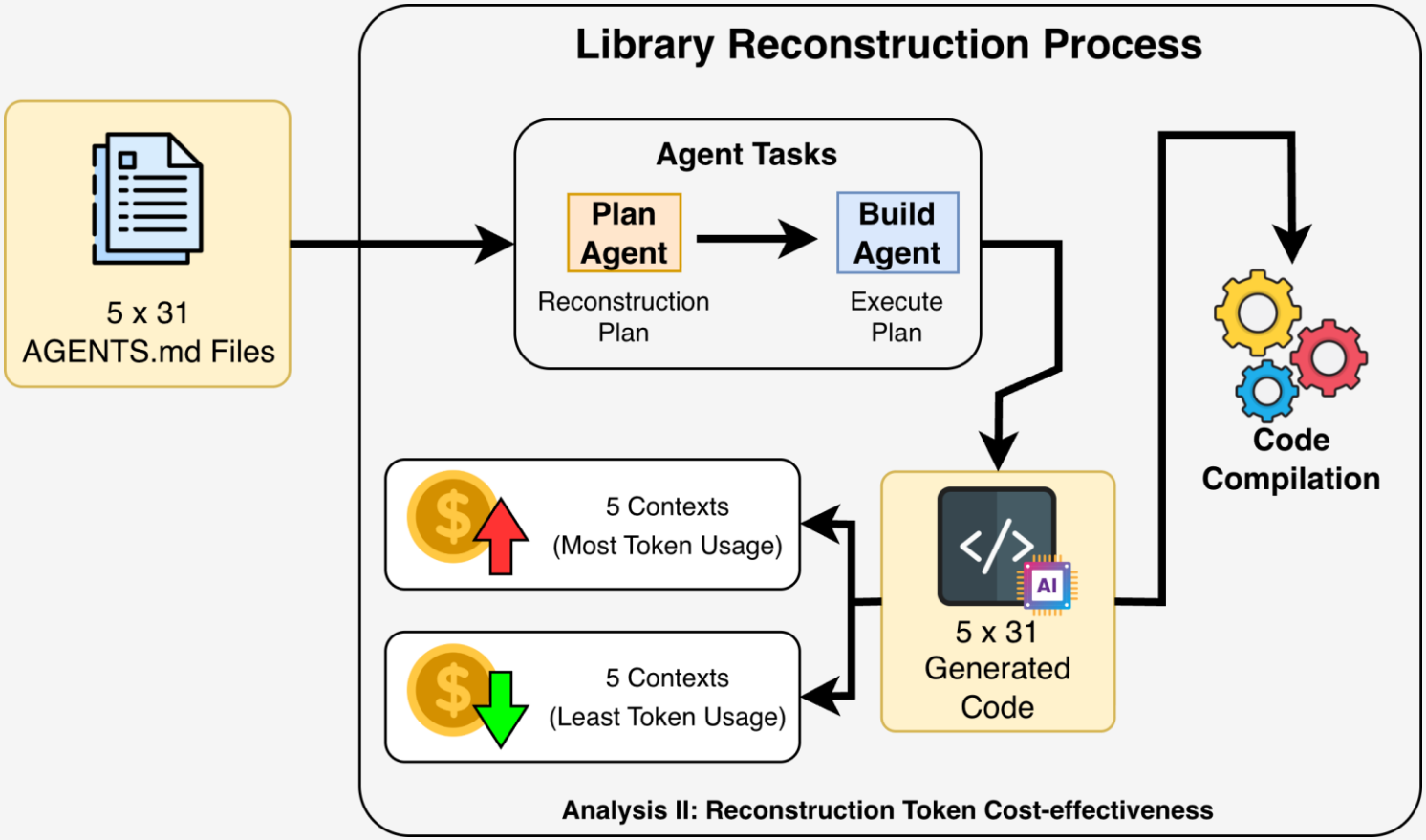


Specification Contexts*:

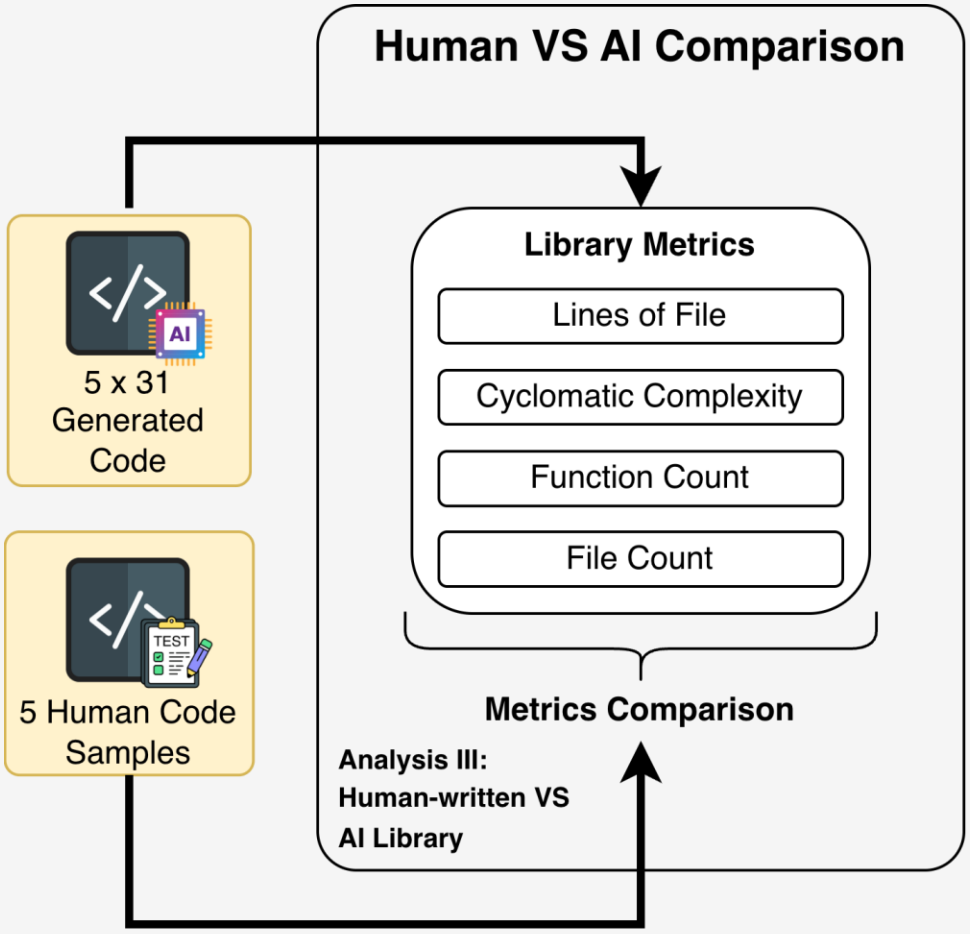
- Repository Overview (RO)
- Function Description (FD)
- Implementation Detail (ID)
- Testing (T)
- Reconstruction Steps (RS)

*Chatlatanagulchai et al., *Agent READMEs: An Empirical Study of Context Files for Agentic Coding*, 2025

Methodology – Library Reconstruction Process



Methodology – Human vs. AI Comparison



Initial Results: RQ1 – Specification Token

TABLE I
TOP AND BOTTOM 5 AVERAGE TOKEN USAGE FOR SPECIFICATION GENERATION BASED ON CONTEXT

Type	Model	Contexts	Token Usage
Top 5	Gemma4	FD-T	24,608.6
		FD-T-RS	23,967.2
		T-RS	23,943.8
		ID-T-RS	23,517.6
		RO-FD-T	23,375.8
	GPT-5.4	FD-T-RS	43,168.4
		FD-T	41,380.8
		FD	40,724.8
		FD-RS	40,581.2
		FD-ID-T-RS	40,345.2
Bottom 5	Gemma4	RO-ID	18,951.8
		RO	19,432.8
		T	20,003.8
		RO-RS	20,008.6
		FD-ID-RS	20,895.8
	GPT-5.4	RO-ID	31,216.4
		RO-T	31,557.8
		ID-T	32,229.6
		RO-ID-RS	32,724.6
		RO-ID-T	33,259.8

RQ1: Which types of context result in the **highest** and **lowest** token usage for creating agentic specification files?

Findings: Token usage cannot be determined based on the amount of context, but rather the level of detail in each context.

Highest: Function Description
Lowest: Repository Overview

Initial Results: RQ2 – Reconstruction Token

TABLE II
TOP AND BOTTOM 5 AVERAGE TOKEN USAGE FOR LIBRARY RECONSTRUCTION BASED ON CONTEXT

Type	Model	Contexts	Token Usage	Compiled
Top 5	Gemma4	RO-FD-ID	30,455.4	80%
		T	27,424.2	100%
		RO-FD-ID-T	27,356.6	60%
		FD-T	25,002.6	100%
		RO-ID	24,819.2	100%
	GPT-5.4	T	44,959.6	100%
		RO-FD-ID-T	40,929.0	100%
		FD-T	40,203.8	100%
		RO-ID-T	39,524.4	100%
		FD-T-RS	39,212.6	100%
Bottom 5	Gemma4	T-RS	18,073.8	100%
		RO-T-RS	18,193.6	100%
		FD-T-RS	18,429.0	60%
		FD-ID-RS	18,605.8	60%
		FD-RS	18,809.8	100%
	GPT-5.4	RO-ID-RS	27,987.0	100%
		RO	28,579.2	100%
		RO-ID	29,207.0	100%
		RS	29,594.6	100%
		RO-T	30,079.0	100%

RQ2: Which contexts **minimize** or **maximize** token usage while still generate code that can be **compiled**?

Findings: Vast majority of the AI-generated library compile successfully across all five library regardless of context.

Highest: Testing
Lowest: Reconstruction Steps

Initial Results: RQ3 – Human vs. AI Comparison

RQ3: How do different specification contexts affect the **structural characteristic** of AI-generated code to the original human-written code?

Findings: In all cases, AI-generated libraries produced significantly fewer files and lower LoF than human-written libraries, regardless of the specification context. They also exhibit a higher maximum CC in 45% of the cases, while producing FC values that are higher in 27% of the cases.

Initial Results: RQ3 – Human vs. AI Comparison

TABLE III
TOP AND BOTTOM 5 FILE COUNT FROM AI-GENERATED LIBRARIES COMPARED TO HUMAN-WRITTEN LIBRARIES

Type	Model	Library	Context	Δ
Top 5	Gemma4	compactor	RO-FD-T	-60.34%
		autolink	ID-RS	-66.67%
		areyoustillthere	ID	-67.57%
		autolink	FD-RS	-69.23%
		areyoustillthere	RO-ID-T	-72.97%
	GPT-5.4	audiodiff	RO-RS	-37.50%
		compactor	RO-FD-RS	-37.93%
		audiodiff	RS	-41.25%
		compactor	ID	-43.10%
		audiodiff	T	-45.00%
Bottom 5	Gemma4	audiodiff	RO-T	-95.00%
		audiodiff	FD	-92.50%
		audiodiff	FD-T	-92.50%
		audiodiff	RO-FD-ID-RS	-92.50%
		autolink	RO-T	-92.31%
	GPT-5.4	audiodiff	FD-T	-90.00%
		audiodiff	ID-T	-90.00%
		audiodiff	RO-ID-T	-90.00%
		audiodiff	FD	-88.75%
		audiodiff	RO-FD	-88.75%

TABLE V
TOP AND BOTTOM 5 LOF FROM AI-GENERATED LIBRARIES COMPARED TO HUMAN-WRITTEN LIBRARIES

Type	Model	Library	Context	Δ
Top 5	Gemma4	autolink	ID-RS	-73.00%
		bitmap	RO-FD-ID	-74.32%
		areyoustillthere	ID	-76.56%
		autolink	RO-FD-RS	-77.65%
		bitmap	RS	-78.70%
	GPT-5.4	areyoustillthere	ID-RS	-16.44%
		audiodiff	T	-59.29%
		areyoustillthere	FD-ID-T-RS	-65.57%
		autolink	RS	-66.29%
		autolink	ID	-66.35%
Bottom 5	Gemma4	audiodiff	FD	-99.45%
		audiodiff	RO-T	-99.42%
		audiodiff	ID-T	-99.35%
		audiodiff	RO-ID-T	-99.22%
		audiodiff	RO-FD-ID-T-RS	-99.19%
	GPT-5.4	audiodiff	RO-FD-T-RS	-98.16%
		audiodiff	RO-FD-ID-T-RS	-97.82%
		audiodiff	ID-T-RS	-97.77%
		audiodiff	RO-ID-T-RS	-97.76%
		audiodiff	RO-FD	-97.70%

Initial Results: RQ3 – Human vs. AI Comparison

TABLE IV
TOP AND BOTTOM 5 FC FROM AI-GENERATED LIBRARIES COMPARED TO HUMAN-WRITTEN LIBRARIES

Type	Model	Library	Context	Δ
Top 5	Gemma4	autolink	T	+28.57%
		areyoustillthere	ID	+27.27%
		areyoustillthere	RO-FD	+18.18%
		autolink	RO-ID-RS	+14.29%
		areyoustillthere	RO	+09.09%
	GPT-5.4	areyoustillthere	ID	+127.27%
		areyoustillthere	FD-T-RS	+109.09%
		areyoustillthere	FD-ID-T-RS	+109.09%
		areyoustillthere	ID-T-RS	+81.82%
		autolink	RS	+71.43%
Bottom 5	Gemma4	compactor	T-RS	-84.17%
		compactor	RO-ID-T	-82.01%
		compactor	RO-FD-ID	-81.29%
		compactor	RO-FD-ID-RS	-80.58%
		compactor	ID-RS	-79.86%
	GPT-5.4	bitmap	ID-T	-50.00%
		bitmap	ID-RS	-50.00%
		bitmap	RO-ID-T	-45.65%
		compactor	FD-ID-RS	-41.73%
		compactor	RO-ID	-41.01%

TABLE VI
TOP AND BOTTOM 5 CC FROM AI-GENERATED LIBRARIES COMPARED TO HUMAN-WRITTEN LIBRARIES

Type	Model	Library	Context	Δ
Top 5	Gemma4	audiodiff	RO-ID	+233.33%
		audiodiff	ID-RS	+216.67%
		audiodiff	RO-FD-ID-RS	+216.67%
		audiodiff	RO-T	+200.00%
		audiodiff	RO-ID-T-RS	+200.00%
	GPT-5.4	audiodiff	T-RS	+383.33%
		audiodiff	RO-RS	+216.67%
		audiodiff	RO-ID-T-RS	+216.67%
		audiodiff	RO	+166.67%
		audiodiff	RO-T	+166.67%
Bottom 5	Gemma4	compactor	T-RS	-77.78%
		compactor	RO-FD-T	-77.78%
		compactor	FD-ID-T-RS	-66.67%
		autolink	RS	-60.00%
		autolink	ID-T	-60.00%
	GPT-5.4	compactor	RO-FD-ID	-33.33%
		compactor	RO-FD-RS	-33.33%
		compactor	RO-ID-RS	-33.33%
		compactor	FD-ID-T-RS	-33.33%
		areyoustillthere	FD-ID	-28.57%

Future Works

- Expand the human-written code dataset
- Verify the correctness of AI-generated code
- Investigate kinds of human interventions needed to refine the correctness of AI-generated code
- Explore more AI-related artifacts: skills.sh, Retrieval Augmented Generation (RAG), memory/context store

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